

Discrete d'Alembertian and Wave Propagation on a Causal Set: A First Numerical Study

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github.com/pikaism/Causal-set-wave

1. Motivation and Goal

This project implements a first working numerical toolkit for exploring causal set phenomenology in 1+1D Minkowski spacetime, built as independent preparation for research in the causal set approach to quantum gravity. The goal was to implement, test, and honestly validate the core kinematical and dynamical ingredients used in this field: sprinkling, the causal order, the discrete Benincasa-Dowker (BD) d'Alembertian, and a scalar field evolution scheme, compared against the known continuum limit. This toolkit is intended as a foundation for more advanced numerical studies of kinematical and dynamical aspects of causal sets, in line with ongoing work in the Theoretical Physics Group at RRI.

2. Methods

2.1 Sprinkling and causal order

N points are generated by sampling exactly N points uniformly in the spacetime volume of a bounded region $R = [0, T] \times [-L, L]$ of 1+1D Minkowski spacetime, with volume $V(R) = 2LT$. This is a fixed-cardinality realization, equivalent to conditioning a genuine Poisson sprinkling (where the point count N would itself be Poisson-distributed with mean $\rho V(R)$) on the total number of points N . This construction is Lorentz-invariant in distribution. Writing $i \leq j$ for "i causally precedes j", the causal order between any two elements is:

$$i \leq j \Leftrightarrow t_j > t_i \text{ and } (t_j - t_i)^2 - (x_j - x_i)^2 \geq 0$$

equivalently, $t_j - t_i \geq |x_j - x_i|$, which is the more intuitive form in 1+1 dimensions.

2.2 Layers and the discrete d'Alembertian

For each element x , define the causal interval between a past element y and x as $I(y, x) = \{z \in C : y \leq z \leq x\}$, and the layer $L_k(x) = \{y \leq x : |I(y, x)| = k\}$. $L_0(x)$ is the set of links (nothing causally between y and x); $L_1(x)$ has exactly one element between them; $L_2(x)$ has exactly two. In 2D (1+1), the Benincasa-Dowker operator, which is retarded (it depends only on $\phi(x)$ and elements in the causal past of x , never the future), takes the form:

$$(B\phi)(x) = (1/l^2) \left[-2\phi(x) + 4\sum_{y \in L_0(x)} \phi(y) - 8\sum_{y \in L_1(x)} \phi(y) + 4\sum_{y \in L_2(x)} \phi(y) \right]$$

where $l = \rho^{-1/2}$, $l^2 = 1/\rho$, and $\rho = N / V(R) = N/(2LT)$ is the sprinkling density. Note that this operator is not a local finite-difference stencil on nearby points: it is a nonlocal, order-based operator whose continuum approximation arises from a carefully weighted combination of layers whose expectation

reproduces the continuum wave operator under suitable conditions. Nonlocality at the discreteness scale is a characteristic feature of causal set field theory, not an artifact of this implementation.

Validation. A naive single-sprinkling test of this operator on simple fields (constant, linear) gave results with large, seemingly inconsistent deviations from the expected value of zero. This was diagnosed, not as an implementation bug, but as a genuine, literature-documented feature: the BD operator's convergence to the continuum d'Alembertian is a statement about the *ensemble average* over many sprinklings, not any single realization [2,3,4]. This was confirmed via ensemble averaging over 30 independent sprinklings (N=500): mean = 0.117, standard error = 1.365, consistent with the theoretical value of 0.

2.3 Wave packet initialization

A Gaussian wave packet is initialized on an early time-slice of the causal set:

$$\begin{aligned}\phi(t_0, x) &= A \cdot \exp(-(x-x_0)^2 / 2\sigma^2) \cdot \cos(k_0(x-x_0)) \\ \partial\phi/\partial t(t_0, x) &= -v \cdot \partial\phi/\partial x(t_0, x), \text{ with } v = 1 \text{ (units } c = 1)\end{aligned}$$

These data specify ϕ and $\partial\phi/\partial t$ on an initial antichain approximating $t=t_0$, defining a standard Cauchy problem for the wave equation. With $v=1$, this transport-type condition exactly matches the initial data of a purely right-moving massless solution $\phi=f(x-t)$ of the continuum wave equation. The discrete evolution scheme, however, does not exactly preserve the massless wave equation once it propagates (Section 2.4), so this exactness applies to the initial data, not to the discrete field at later times.

2.4 Field evolution

The field is evolved by processing elements in increasing time order and solving the discrete field equation $(B\phi)(x) - m^2\phi(x) = 0$ for the one unknown $\phi(x)$, using only already-known values in its causal past. This is possible because the BD operator is retarded: $B\phi(x)$ depends on $\phi(x)$ itself plus values already known from the causal past. This produces a retarded algebraic recursion; it should not be interpreted as a rigorously stable discrete Cauchy evolution of a hyperbolic PDE, and the project tests the numerical behavior of this particular realization of the operator rather than claiming a general evolution scheme. The present simulations use the massless case $m=0$ throughout, so the sign convention for a mass term does not affect the reported results; for the $(+,-)$ signature used here, the massive continuum equation would be $\partial_t^2\phi - \partial_x^2\phi + m^2\phi = 0$. The retarded-solve idea is standard in the causal set literature (see e.g. Johnston [5] on causal set field propagators), though the present damped recursion is a specific numerical realization and should not be read as reproducing Johnston's formal propagator construction.

Instability found and fixed. A naive exact solve of this equation was found to be numerically unstable for the finite causal sets and parameters studied here: the field magnitude grew exponentially (from $O(1)$ to $O(10^{25})$) over a handful of time bins, even though the direction of propagation was qualitatively correct. Unlike the regular-grid leapfrog scheme (Section 2.5), this recursion has no evident CFL-like stability parameter controlling the amplification of perturbations, so noise is amplified multiplicatively across the many "generations" of causal depth in the set. The fix applied was to (a) replace layer sums with layer *averages* (which changes the operator's normalization, and is therefore no longer the BD

operator itself, in addition to being a distinct choice from the subsequent under-relaxation), bounding the update regardless of layer size, and (b) apply an under-relaxation / damping factor α (used here: $\alpha=0.3$) on top of that. This is a standard numerical stabilization technique, applied here at the acknowledged cost of no longer solving the exact discrete field equation.

2.5 Continuum reference solution

A standard explicit leapfrog finite-difference scheme solves the same massless wave equation $\partial_t^2 \phi - \partial_x^2 \phi = 0$ on a regular grid, with identical initial data, for direct comparison:

$$\phi_j^{n+1} = 2\phi_j^n - \phi_j^{n-1} + r^2(\phi_{j+1}^n - 2\phi_j^n + \phi_{j-1}^n), \quad r = \Delta t / \Delta x$$

Stability of this explicit scheme requires $r \leq 1$ (the CFL condition); the implementation used here sets $r = 0.5$, well within the stable range. This is the small parameter, absent in the discrete causal-set recursion, that makes the continuum reference numerically well behaved.

2.6 The n-interval spectrum

Following Surya [6], the n-interval spectrum is defined as $N_m = \#\{(y,x) : y \leq x, |I(y,x)| = m\}$, the number of causally related pairs whose causal interval has cardinality m , counted over the whole causal set. This order-invariant "fingerprint" was computed as an additional diagnostic of continuumlike behavior.

3. Results

3.1 Wave packet propagation: discrete vs continuum

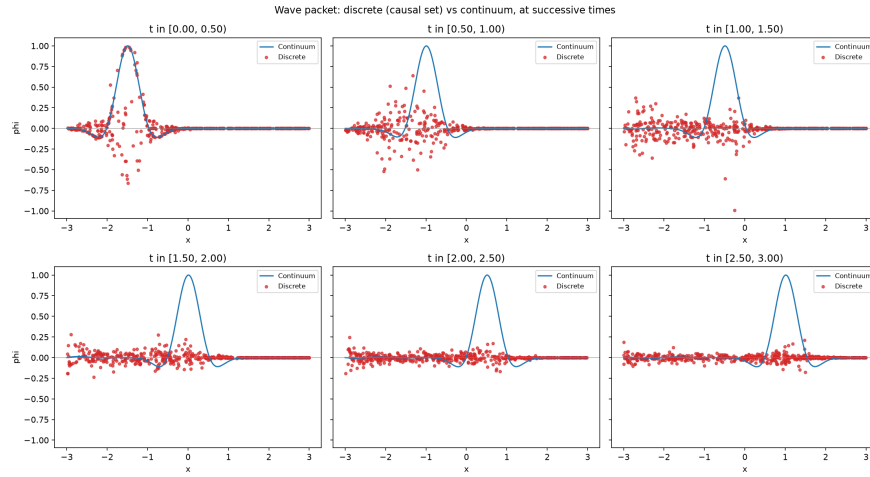


Figure 1. Discrete (causal set, red) vs continuum (blue) field profile at six successive time windows.

At the initial time bin, the discrete (red) and continuum (blue) solutions agree closely, as expected since both share the same initial data. At later times, the continuum solution retains a clean, moving Gaussian shape, while the discrete solution becomes visibly noisier, though it broadly tracks the same rightward propagation. Quantitatively, the energy-weighted centroid in the discrete case drifts rightward but more slowly and with larger fluctuations than in the continuum case, reflecting the dissipative effect of the damping factor $\alpha=0.3$. This is a direct, visual and quantitative confirmation that the damping needed to stabilize the discrete evolution measurably degrades the coherence of the propagating wave packet relative to the continuum — a real, quantified limitation of the current scheme.

3.2 Convergence with density

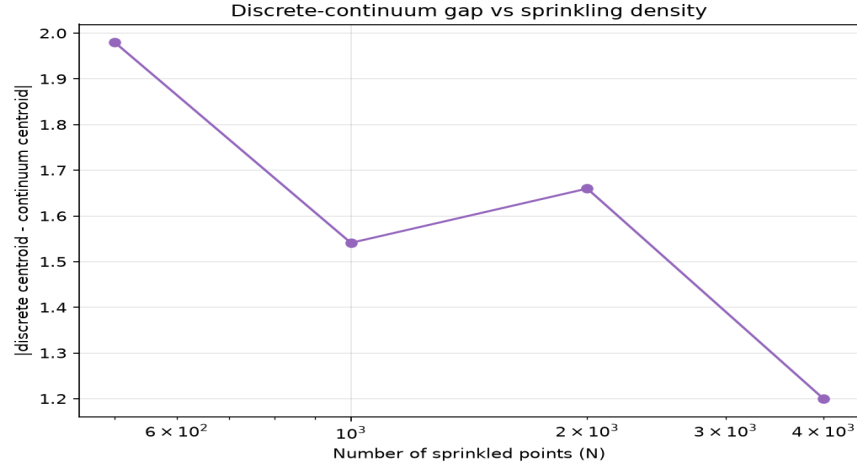


Figure 2. Gap between discrete and continuum centroids at the final time bin, as a function of sprinkling density N .

Comparing the discrete and continuum energy-weighted centroids at the final time bin, across increasing sprinkling density ($N = 500, 1000, 2000, 4000$), shows an overall decreasing trend in the gap between them, consistent with (though not a clean demonstration of) convergence to the continuum limit as density increases. Given the known per-realization variance of the BD operator and the modest number of trials, this should be regarded as a qualitative indication of convergence rather than a precise scaling measurement.

3.3 n-interval spectrum

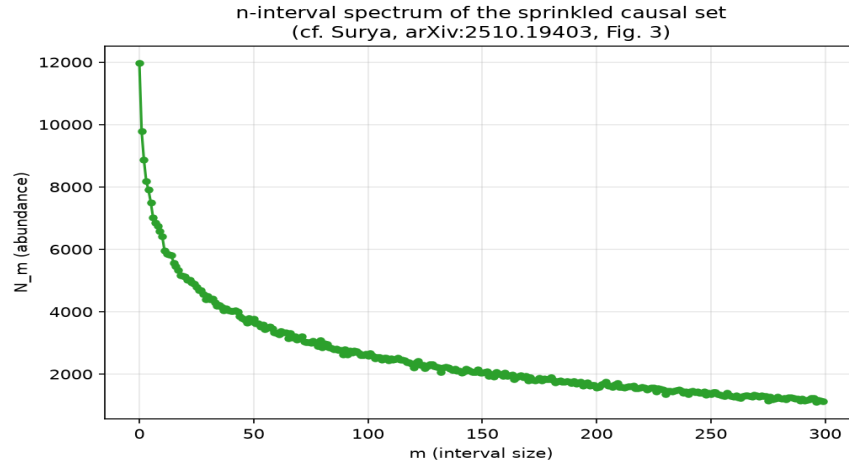


Figure 3. n -interval spectrum N_m of the sprinkled causal set ($N=2000$), cf. Surya [6] Fig. 3.

The computed spectrum N_m decreases smoothly and monotonically with m , matching the qualitative "continuumlike" signature described in Surya [6] (Fig. 3) for a causal diamond in Minkowski spacetime. A smooth, monotonically decreasing N_m is characteristic of sprinklings into continuum-like geometries; significant deviations from this shape have been associated with non-continuumlike causal sets. This provides an independent, order-theoretic diagnostic consistent with the sprinkled causal set's continuum-like origin, using a measure directly drawn from current causal set research — though on its own it is a diagnostic, not a complete proof of continuum approximation.

4. Limitations

- `compute_layers` / `interval_size` scale poorly (roughly $O(N^2)$ to $O(N^3)$ depending on causal set density), making large- N or many-trial ensemble studies slow; a full, statistically rigorous convergence study would require an optimized implementation (e.g. time-sorted indexing to avoid re-scanning the full past for every interval check).
- The discrete evolution scheme requires damping ($\alpha=0.3$) and a switch from layer sums to layer averages to remain numerically stable; both changes alter the operator's normalization and mean it no longer solves the exact BD field equation, and measurably slow the effective propagation speed relative to the continuum solution.
- The BD operator has large single-realization variance; only ensemble-averaged behavior matches the continuum expectation cleanly.
- The retarded algebraic solve used for evolution is not a rigorously stable discrete Cauchy evolution of a hyperbolic PDE; it is a specific numerical realization whose behavior was tested empirically, not a general-purpose evolution scheme.
- Only 1+1D Minkowski spacetime is implemented.

5. What I would do differently with more time

- Optimize `interval_size/compute_layers` (e.g. via time-sorted binary search or spatial indexing) to enable a proper, statistically significant convergence study across many trials and higher densities.
- Investigate a better-normalized, less lossy stabilization scheme for the discrete evolution than simple under-relaxation on layer averages, potentially drawing on retarded Green's function constructions on causal sets (e.g. Johnston [5]) or alternative discrete wave operators that preserve the BD normalization exactly.
- Extend to 2+1D Minkowski spacetime.
- Extend the n -interval spectrum analysis to compare multiple spacetime regions (e.g. causal diamond vs hypercube), following the closeness-function construction in Surya [6], and explore its sensitivity to dimension and curvature.

References

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